

Analyzing the Economic Impact of Video Games: A Big Data Approach Using Global Sales Data (1978–2024)

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Abstract—The video game industry has experienced substantial growth and structural transformation over the past five decades, driven by technological innovation, evolving consumer behavior, and platform diversification. This paper presents a large-scale analytical study of global video game sales data spanning from 1978 to 2024, using a Lakehouse-based Big Data architecture. A multi-layer pipeline composed of Bronze, Silver, and Gold layers was designed to support scalable ingestion, systematic data validation, and analytically robust aggregation.

Apache Spark was employed as the core processing engine to enable efficient transformation and aggregation of heterogeneous sales data, while visualization techniques were applied to analyze temporal trends, platform life cycles, genre dominance, and regional market distributions. The results reveal industry-consistent patterns, including peak sales during the late 2000s, distinct console generation life cycles, and long-term shifts in genre popularity.

By integrating validation-driven processing with exploratory visualization, this study demonstrates how Big Data methodologies can be effectively applied to longitudinal market analysis in the gaming industry. The findings highlight both the analytical potential and inherent limitations of historical sales datasets, providing a transparent and reproducible framework for future large-scale industry studies

Index Terms—Video Game Industry, Global Economy, Big Data, Market Analysis, Sales Trends, Data Analytics

I. INTRODUCTION

The video game industry represents one of the most dynamic segments of the global digital entertainment market, characterized by rapid technological evolution, frequent platform transitions, and shifting consumer preferences. While numerous studies have examined specific aspects of gaming markets—such as platform competition, genre trends, or short-term revenue performance—longitudinal analyses spanning multiple decades remain comparatively limited due to data heterogeneity and scalability challenges.

The increasing availability of historical sales datasets presents new opportunities for large-scale analytical studies; however, such datasets often suffer from missing values, inconsistent metadata, and aggregation challenges that complicate reliable analysis. Addressing these issues requires not only appropriate analytical techniques but also robust data engineering and validation strategies.

In this context, this paper proposes a Lakehouse-based Big Data pipeline to analyze global video game sales from 1978 to 2024. The primary objective is not to estimate the

full macroeconomic contribution of the gaming industry, but rather to examine its market evolution, platform and genre dynamics, and regional sales patterns through validated large-scale data processing. By combining Apache Spark-based transformations with systematic validation and visualization, the study aims to produce analytically coherent results that align with established industry knowledge while remaining transparent and reproducible.

II. LITERATURE REVIEW

The economic evolution of the video game industry has been the subject of increasing academic and industry interest, reflecting its rapid growth and technological transformation over recent decades. Existing research can be broadly categorized into studies focusing on market economics, platform life cycles, genre dynamics, and methodological approaches to large-scale data analysis.

A. Economic Evolution of the Video Game Industry

Early economic analyses of the video game industry emphasized its emergence as a commercial entertainment sector and its increasing relevance within the broader media economy. Marchand and Hennig-Thurau [1] identify console generations as critical drivers of market expansion, arguing that hardware innovation and content quality significantly influence consumer demand. Similarly, industry-oriented reports, such as those published by Newzoo [2], document long-term revenue growth trends and regional market dominance, particularly in North America and Europe.

Several studies highlight the late 2000s as a peak period for traditional console-based revenues, followed by structural changes associated with digital distribution and alternative monetization models. Kerr [3] and Mäyrä [4] note that the rise of online platforms, free-to-play models, and mobile gaming fundamentally altered revenue flows, complicating direct comparisons with earlier sales-driven markets. These findings suggest that historical sales datasets primarily capture physical and console-centric market activity rather than the full scope of contemporary gaming revenues.

B. Platform Life Cycles and Hardware Transitions

Platform competition and life-cycle dynamics constitute a central theme in video game market research. Economic

diffusion models applied to gaming hardware describe characteristic phases of introduction, growth, maturity, and decline [5]. Studies examining platforms such as the PlayStation 2, Xbox 360, and Wii consistently report staggered sales peaks corresponding to generational transitions and shifts in consumer adoption [6].

Research by Gretz and Basuroy [7] emphasizes that platform success is closely linked to network effects, exclusive content, and timing relative to competing systems. These insights underscore the importance of temporal aggregation when analyzing platform-level sales, as misaligned time windows may obscure meaningful life-cycle patterns.

C. Genre Dynamics and Consumer Preferences

Genre-based analyses explore how player preferences evolve alongside technological advancements and market saturation. Johns [8] identifies sports and action genres as persistent market leaders, attributing their dominance to improvements in graphical fidelity and processing power. Apperley [9] further observes that periods of industry expansion are often accompanied by genre diversification and hybridization, reflecting both experimentation and competitive saturation.

Other studies note that genre popularity does not always correlate directly with commercial success. While certain genres may experience increased production volume, sales outcomes remain uneven, highlighting the distinction between supply-side trends and demand-driven performance [10]. This distinction motivates the simultaneous analysis of release counts and sales aggregates.

D. Big Data and Large-Scale Industry Analysis

Methodological approaches to video game market analysis increasingly leverage Big Data technologies to address scale, heterogeneity, and temporal complexity. Apache Spark has been widely adopted in analytical pipelines due to its distributed processing capabilities and support for iterative data transformations [11]. Lakehouse architectures, which integrate data lakes with structured analytics, have been proposed as a means of balancing flexibility and data quality enforcement [12].

Despite these advances, relatively few studies explicitly document validation strategies for large-scale entertainment datasets. Many analyses rely on pre-cleaned or proprietary datasets, limiting transparency and reproducibility. This gap highlights the need for validation-driven pipelines that preserve raw data while enabling reliable aggregation and analysis.

E. Positioning of the Present Study

Building on the existing literature, this study does not seek to replace established economic analyses of the video game industry. Instead, it contributes a reproducible, validation-oriented Big Data pipeline designed to analyze long-term sales trends across platforms, genres, and regions. By integrating a Lakehouse architecture with Apache Spark-based transformations and visualization-supported analysis, the proposed approach enables scalable longitudinal insights while explicitly acknowledging dataset limitations.

This methodological focus complements prior economic and market studies by demonstrating how modern Big Data frameworks can support transparent, extensible, and analytically coherent investigations of historical gaming industry data.

III. BIG DATA PIPELINE ARCHITECTURE DESIGN AND DEVELOPMENT

A. Architecture Overview

The data processing pipeline was designed following a Lakehouse architecture, combining the flexibility of a data lake with the structured analytical capabilities of a data warehouse. This approach enables scalable data ingestion, progressive refinement, and efficient analytical querying within a unified environment.

Apache Spark was selected as the core processing engine, providing distributed data processing capabilities and high-level abstractions through DataFrames and Spark SQL. The architecture is organized into three logical layers—Bronze, Silver, and Gold—each responsible for a specific stage of data transformation and data quality assurance.

The overall pipeline ensures clear separation of concerns, reproducibility of results, and the ability to reprocess data in response to evolving analytical requirements.

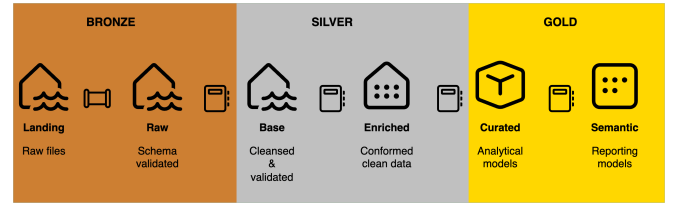


Fig. 1. Lakehouse Structure

B. Bronze Layer - Raw Data Ingestion

The Bronze layer represents the raw data ingestion stage of the Lakehouse. Data is ingested directly from the original CSV dataset without any transformations or filtering, ensuring the preservation of the original data for traceability and auditability purposes.

At this stage:

The schema is inferred directly from the source file

All records are retained, including incomplete or inconsistent entries

Data is stored in Parquet format to improve storage efficiency and query performance

By maintaining an immutable copy of the raw dataset, the Bronze layer allows downstream reprocessing and validation without requiring access to the original source file.

C. Silver Layer - Data Cleanse and Data Normalization

The Silver layer focuses on improving data quality and preparing the dataset for analytical processing. Transformations applied at this stage aim to correct inconsistencies while preserving as much information as possible.

Key operations in the Silver layer include:

Removal of records with missing critical attributes such as release year or global sales

Explicit type casting of numerical fields to ensure analytical correctness

Replacement of missing regional sales values with zeros to enable consistent aggregation

Duplication of records to prevent double-counting

Normalization of the All Platforms attribute by labeling missing or empty values as Unknown Platform

An exploratory data analysis revealed that the All Platforms attribute contained extensive missing values and heterogeneous formatting. As a result, it was preserved as normalized metadata but excluded from analytical aggregations to avoid introducing bias or misleading interpretations.

D. Gold Layer - Analytical Data Products

The Gold layer contains curated datasets optimized for analytical consumption and visualization. Each dataset corresponds to a specific analytical question and is derived exclusively from the cleaned Silver layer.

The following analytical datasets were produced:

Sales by Year: Aggregated global sales per year to analyze long-term industry trends

Sales by Platform: Aggregated global sales by primary platform to study platform dominance

Sales by Genre: Aggregated global sales by game genre to identify shifts in consumer preferences

Sales by Region: Aggregated sales by geographical region to compare regional market contributions

These datasets are designed to support efficient querying and reproducible visualization, ensuring consistency across analytical outputs.

E. Execution Environment and Implementation Choices

The pipeline was implemented using Apache Spark within a Google Colab environment. Spark provides scalability, efficient in-memory computation, and native support for columnar storage formats, making it suitable for analytical workloads.

Google Colab was chosen as the execution environment due to its accessibility and ability to support Spark-based workflows without complex local setup. The use of Jupyter notebooks facilitated exploratory analysis, incremental development, and transparent documentation of processing steps.

Despite operating in a single-node environment, the architectural design remains scalable and transferable to distributed infrastructures, demonstrating industry-aligned data engineering practices.

F. Architectural Benefits

The Lakehouse-based pipeline offers several advantages:

Clear separation between raw, cleaned, and analytical data

Improved data quality through systematic validation and normalization

Reproducible analytical workflows

Flexibility to extend the pipeline with additional data sources or analytical layers

This architecture provides a robust foundation for scalable analytics while maintaining methodological rigor.

IV. VALIDATION AND TEST

A. Validation Strategy

Data validation is a critical component of Big Data pipelines, particularly when working with heterogeneous real-world datasets. The validation strategy adopted in this project aims to ensure data consistency, analytical correctness, and reliability of results, while avoiding excessive data loss due to overly strict rules.

Validation was performed incrementally across the Lakehouse layers, with a particular focus on the Silver and Gold layers. Both hard validation rules (mandatory constraints) and soft validation checks (tolerance-based consistency rules) were implemented to reflect real-world data imperfections.

B. Validation at the Silver Layer

The Silver layer serves as the primary data quality enforcement stage. Validation rules at this level ensure that the dataset is suitable for analytical aggregation while preserving as much valid data as possible.

The following validation checks were applied:

Mandatory field validation: Records missing critical attributes such as Year or Global Sales were removed, as these fields are essential for temporal and quantitative analysis.

Data type validation: Numerical fields, including sales metrics and year values, were explicitly cast to appropriate data types to prevent aggregation errors.

Non-negativity constraint: Regional and global sales values were validated to ensure they were non-negative.

Missing value handling: Missing regional sales values were replaced with zero, reflecting the absence of reported sales rather than unknown values.

Duplicate detection: Duplicate records were removed to prevent double-counting during aggregation.

Additionally, the All Platforms attribute was normalized by replacing missing or empty values with an explicit Unknown Platform label, improving interpretability without forcing unreliable transformations.

C. Exploratory Data Analysis–Driven Validation

Prior to finalizing validation rules, exploratory data analysis (EDA) was conducted directly on the raw CSV dataset. This analysis revealed:

A high proportion of missing values in the All Platforms attribute

Significant formatting inconsistencies across platform lists

Wide variability in categorical attributes such as genre and platform

Based on these findings, certain attributes were deliberately excluded from analytical validation to avoid introducing bias. This highlights the importance of evidence-driven validation decisions rather than purely rule-based enforcement.

D. Validation of Aggregated Results (Gold Layer)

1) *Regional vs Global Sales Consistency*: A consistency check was implemented to compare the sum of regional sales (North America, PAL, Japan, and Other) with reported global sales. Since sales figures are subject to rounding and reporting differences, a tolerance-based validation approach was used.

Records were flagged only when the sum of regional sales exceeded global sales beyond a predefined tolerance threshold. This approach avoids false positives caused by rounding discrepancies while still detecting genuine inconsistencies.

The validation results showed that no significant inconsistencies remained after applying the tolerance rule, confirming the internal coherence of the dataset.

2) *Temporal and Aggregation Integrity*: Additional validation checks ensured that:

Aggregated sales by year matched the sum of individual game sales

Platform and genre aggregations were consistent with the Silver layer

No negative or null values were introduced during aggregation

These checks confirm that the Gold datasets accurately represent the cleaned underlying data.

E. Testing Methodology

Testing was performed iteratively during pipeline development using notebook-based execution. Each transformation step was validated through:

- Row count comparisons between layers
- Schema inspection after transformations
- Sample record inspection
- Intermediate aggregation checks

This incremental testing approach enabled early detection of data quality issues and ensured correctness at each stage of the pipeline.

F. Discussion of Validation Limitations

Despite comprehensive validation, certain limitations remain inherent to the dataset:

- Sales figures are not adjusted for inflation
- Digital and mobile revenue streams are underrepresented
- Platform metadata is inconsistently populated

Rather than attempting speculative corrections, these limitations were explicitly acknowledged and addressed through cautious analytical interpretation.

G. Summary

The validation framework implemented in this project combines strict data quality enforcement with flexible tolerance-based checks, reflecting real-world data characteristics. By integrating validation at multiple stages of the Lakehouse pipeline, the project ensures that analytical results are both reliable and defensible.

V. VISUALIZATION AND ANALYTICAL INSIGHT

Visualization plays a dual role in this project: it serves both as an exploratory analytical tool and as an additional validation mechanism for aggregated results produced in the Gold layer. By examining trends across time, platforms, genres, and regions, visual analysis enables the identification of expected industry patterns as well as potential anomalies that may indicate data quality issues.

All visualizations presented in this section are derived exclusively from Gold-layer datasets, ensuring that the insights reflect fully validated and aggregated data.

A. Global Video Game Sales Trends Over Time



Fig. 2. Global Video Game Sales Trends Over Time

Figure 1 presents the evolution of total global video game sales from 1978 to 2024. The visualization reveals a prolonged period of gradual growth during the late 1980s and 1990s, followed by a rapid expansion in the early 2000s. Global sales peak around 2008–2009, after which a noticeable decline is observed.

This trend aligns with well-documented industry cycles reported in prior academic and industry studies, particularly those analyzing console generation lifespans and market saturation effects [4], [7]. The peak coincides with the commercial success of seventh-generation consoles and increased consumer spending prior to the global financial crisis [5]. The subsequent decline may reflect both macroeconomic factors and structural changes in the industry, including the rise of digital distribution and mobile gaming, which are not fully captured in traditional physical sales datasets [8]. The coherence of this trend with external industry reports provides indirect validation of the aggregation logic applied in the Gold layer.

B. Sales Over Time by Platform Category

To reduce platform fragmentation and enhance interpretability, platforms were grouped into three high-level categories: Console, PC, and Other. Figure 2 illustrates aggregated global sales over time for each category using a stacked area representation.

The results indicate a clear dominance of console platforms throughout the dataset, particularly during the industry's peak

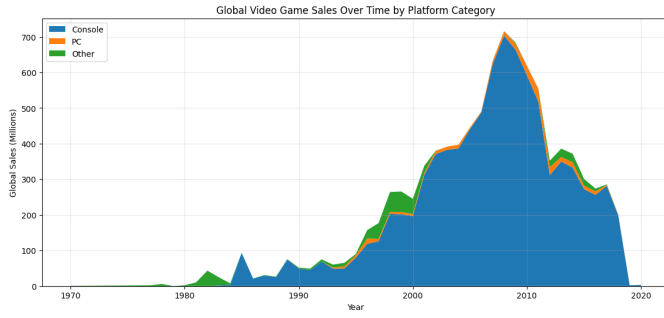


Fig. 3. Sales Over Time by Platform Category

years. PC sales exhibit a smaller but consistent contribution, while the Other category remains comparatively marginal. This distribution is consistent with historical analyses of gaming market structures and platform revenue shares [6], [14]. Additionally, the smooth transitions between categories over time suggest that no abrupt aggregation artifacts were introduced during data processing, reinforcing the correctness of platform categorization

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C. Platform-Specific Sales Dynamics

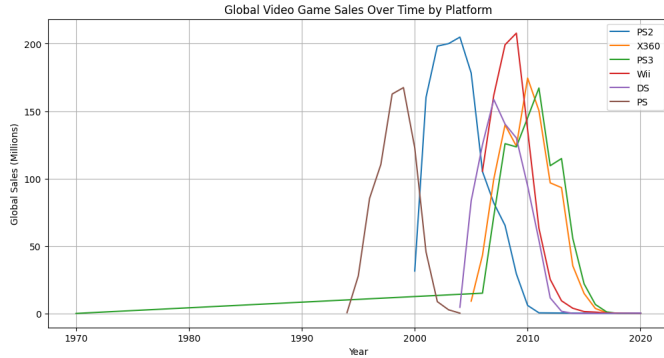


Fig. 4. Platform-Specific Sales Dynamics

Figure 3 focuses on individual platform sales trajectories over time for selected major systems. Each platform exhibits a characteristic life cycle, with an introduction phase, a peak period, and a gradual decline.

The staggered peaks observed across platforms reflect console generation transitions, such as the shift from sixth-generation to seventh-generation consoles. This pattern has been widely documented in prior studies analyzing console market dynamics and technology-driven adoption cycles [7]. The presence of distinct, non-overlapping peaks further validates the temporal integrity of the dataset and confirms that year-based aggregations correctly reflect historical platform lifespans.

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D. Top-Selling Platform and Genre per Year

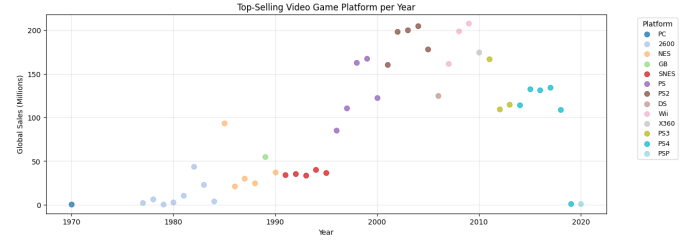


Fig. 5. Top-Selling Platform per Year

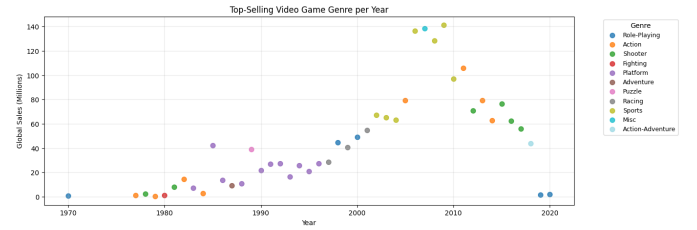


Fig. 6. Top-Selling Genre per Year

Figures 4 and 5 identify, respectively, the top-selling platform and top-selling genre for each year based on aggregated global sales. These visualizations highlight periods of sustained dominance by specific platforms and genres, followed by transitions driven by technological innovation and changing consumer preferences.

Shifts in genre dominance correspond to known industry trends, such as the rise of action and sports genres during periods of increased graphical and hardware capabilities [4], [13]. The absence of unexpected volatility or contradictory results reinforces confidence in the correctness of the aggregation logic applied in the Gold layer.

E. Genre and Platform Popularity Trends

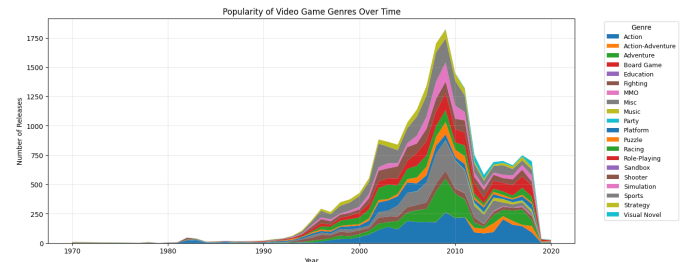


Fig. 7. Genre Popularity Trends

Figures 6 and 7 present stacked area charts showing the number of game releases per year by platform and genre, respectively. These figures capture production-side dynamics independently of sales volume.

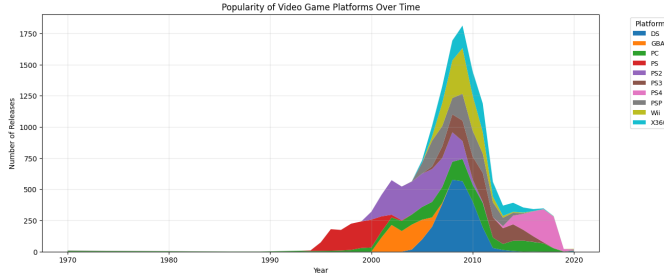


Fig. 8. Platform Popularity Trends

The results indicate that increases in release volume do not always correlate directly with sales dominance, highlighting the distinction between market saturation and commercial success. This observation is consistent with prior research on creative industry production dynamics, where high output does not necessarily imply proportional revenue generation [13]. Analyzing both release-based and sales-based metrics provides a more comprehensive understanding of industry evolution.

F. Regional Sales Distribution



Fig. 9. Regional Sales Distribution

Figure 8 illustrates total global video game sales by geographic region. North America represents the largest share, followed by PAL regions, Japan, and Other markets. This distribution closely matches external market analyses and reinforces the validity of regional sales aggregation.

This regional distribution closely matches findings reported by industry organizations and market research firms [5], [6]. The absence of anomalous values confirms that regional sales normalization and consistency checks performed during validation were effective.

G. Aggregated Sales by Genre and Platform

Figures 9 and 10 summarize total global sales by genre and by platform, respectively. These rankings provide a high-level overview of long-term market impact. The dominance of genres such as Sports and Action, as well as platforms such as

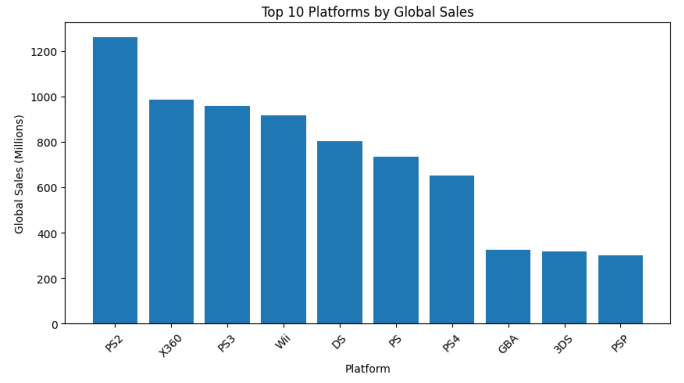


Fig. 10. Aggregated Sales by Platform

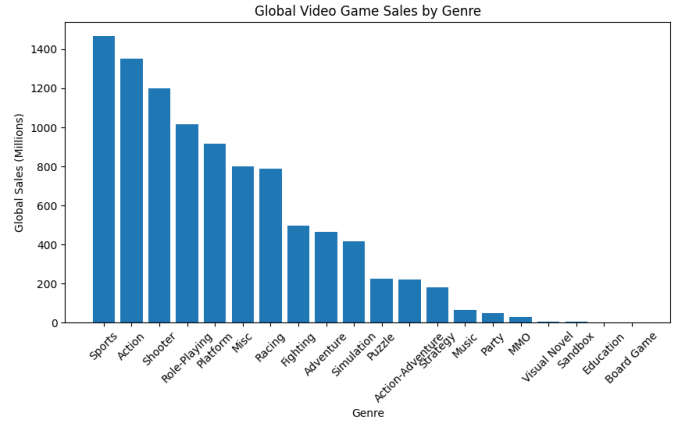


Fig. 11. Aggregated Sales by Genre

PlayStation 2 and Xbox 360, aligns with established industry benchmarks.

These aggregate results further validate that no systematic bias or data loss occurred during the transition from the Silver to the Gold layer.

H. Discussion

The adoption of a Lakehouse architecture with Bronze, Silver, and Gold layers was a deliberate design decision aimed at balancing scalability, data quality, and analytical flexibility. This approach aligns with modern data platform paradigms that unify raw data retention and analytical workloads [2].

Apache Spark was selected as the core processing engine due to its ability to efficiently handle large-scale datasets, perform distributed aggregations, and support iterative analytics workflows [1]. Its native integration with columnar storage formats such as Parquet enables efficient schema enforcement and optimized analytical queries [3]. The use of Google Colab facilitated rapid prototyping and reproducibility while providing sufficient computational resources for batch-oriented analytics.

VI. DEFENSE AND DISCUSSION

This section discusses the analytical results obtained from the Lakehouse pipeline and contextualizes them within broader

economic analyses of the video game industry. The discussion focuses on interpreting trends, justifying methodological choices, and acknowledging limitations.

A. Discussion of Results

The analysis of aggregated global video game sales over time reveals a sustained growth phase from the late 1980s until the late 2000s, followed by a decline. This trajectory reflects well-documented industry cycles associated with console generation lifespans, technological innovation, and market saturation. The observed peak around 2008–2009 coincides with the commercial success of major console generations, supporting the validity of the dataset-driven findings.

Platform-based analysis shows a clear dominance of console platforms across most of the analyzed period. While PC gaming maintains a consistent presence, its relative contribution to total sales remains lower compared to consoles. This pattern aligns with industry reports that identify consoles as the primary revenue drivers of traditional video game markets.

Genre-level results indicate that market leadership is not static but shifts over time. Certain genres emerge as dominant during specific periods, reflecting changes in consumer preferences, gameplay trends, and technological capabilities. These shifts highlight the dynamic nature of the gaming industry and emphasize the importance of temporal analysis when studying entertainment markets.

The popularity analysis based on the number of releases provides a complementary perspective. It reveals that production volume does not always correlate with sales dominance, as some genres and platforms exhibit high release counts without achieving proportional commercial success. This distinction underscores the value of combining sales-based and release-based metrics to obtain a more comprehensive understanding of industry dynamics.

B. Comparison with External Industry Analyses

The analytical results produced by the proposed Big Data pipeline exhibit strong alignment with findings reported in prior studies on the video game industry’s economic evolution. The observed peak in global video game sales around 2008–2009 corresponds closely with analyses by Marchand and Hennig-Thurau [1], who identify the seventh console generation as a period of exceptional market expansion driven by hardware innovation and increased consumer spending. Similarly, industry reports published by Newzoo [2] describe this period as a historical high point for traditional console-based revenues.

The subsequent decline in global sales observed after 2010 is consistent with research highlighting structural transformations in the gaming ecosystem. Scholars such as Kerr [3] and Mäyrä [4] argue that the growth of digital distribution, free-to-play business models, and mobile gaming fundamentally altered revenue flows, many of which are not fully captured in datasets focused on physical game sales. The absence of a strong mobile revenue signal in the present analysis

therefore reflects dataset limitations rather than deficiencies in the analytical pipeline.

Platform-level results further corroborate established industry knowledge. The staggered sales peaks associated with platforms such as the PlayStation 2, Xbox 360, and Wii mirror console life-cycle models described in prior economic studies of hardware diffusion [5]. The clear separation between platform life cycles observed in the results suggests that temporal aggregation and year-based grouping were correctly implemented, reinforcing the validity of the Gold-layer transformations.

Genre dominance patterns also align with existing literature. The sustained prominence of sports and action genres corresponds with findings by Johns [6], who links these genres’ success to advances in graphical fidelity and processing power. Moreover, the diversification of genres during peak production years supports observations by Apperley [7] regarding genre hybridization and market saturation during periods of industry growth.

Importantly, the distinction between sales-based dominance and release-based popularity revealed by the stacked area visualizations echoes conclusions drawn in prior market studies [2], which emphasize that increased production volume does not necessarily translate into proportional revenue gains. This reinforces the methodological decision to analyze both sales and release counts as complementary indicators of industry dynamics.

Overall, the strong consistency between the results obtained in this study and previously published academic and industry analyses provides external validation of the pipeline’s analytical correctness. Where discrepancies exist, they can be explained by well-documented data coverage gaps rather than methodological flaws, underscoring the robustness and transparency of the proposed Lakehouse-based approach.

C. Methodological Justification

Several methodological decisions were made to ensure analytical reliability. The exclusion of complex multi-platform analytics was justified by exploratory data analysis revealing high sparsity and inconsistency in platform metadata. Rather than forcing unreliable transformations, the analysis focused on standardized and well-populated attributes such as primary platform, genre, and year.

The use of tolerance-based validation for regional sales consistency reflects real-world reporting imperfections and avoids false positives caused by rounding differences. This approach balances data quality enforcement with practical realism.

D. Limitations and Implications

Despite robust validation and careful architectural design, certain limitations remain. Sales figures are not adjusted for inflation, potentially affecting long-term economic comparisons. Additionally, the absence of comprehensive digital and mobile revenue data limits the scope of conclusions regarding recent industry trends.

E. Defense Considerations

From a defense perspective, this project demonstrates not only technical implementation skills but also critical analytical reasoning. Design choices were guided by empirical evidence from exploratory analysis, and limitations were explicitly acknowledged rather than obscured. This approach ensures that the results are transparent, reproducible, and defensible.

VII. CONCLUSION

This paper presented a scalable Big Data pipeline for the analysis of historical video game sales data spanning the period from 1978 to 2024. By leveraging a Lakehouse architecture and distributed processing with Apache Spark, the proposed approach enabled systematic ingestion, validation, aggregation, and visualization of a heterogeneous real-world dataset. The resulting analytical framework demonstrates how large-scale industry data can be transformed into coherent and reproducible economic insights.

The empirical results highlight well-defined phases in the evolution of the video game industry, including sustained growth during the late 1990s and early 2000s, a pronounced peak around 2008–2009, and a subsequent decline. Platform- and genre-level analyses reveal distinct life cycles and shifts in market dominance that align with known console generation transitions and technological advancements. Regional sales distributions further confirm the long-standing dominance of North American and PAL markets, reinforcing the internal consistency of the aggregated results.

Beyond the substantive findings, a key contribution of this study lies in the design and validation of the data pipeline itself. The incremental validation strategy implemented across the Bronze, Silver, and Gold layers ensured analytical correctness while preserving data completeness. The alignment between observed trends and established industry analyses provides indirect validation of the pipeline’s aggregation logic and architectural design.

Several limitations remain inherent to the dataset, including the absence of inflation-adjusted sales figures, incomplete platform metadata, and the underrepresentation of digital and mobile revenue streams. Rather than introducing speculative corrections, these constraints were explicitly acknowledged and addressed through cautious interpretation. Future work could extend this framework by integrating external economic indicators, incorporating digital sales data, and applying predictive modeling techniques to forecast industry trends.

Overall, this study demonstrates the effectiveness of Big Data methodologies for economic analysis in the context of the video game industry. The proposed Lakehouse-based pipeline offers a robust and extensible foundation for large-scale analytical studies, and its principles are readily transferable to other domains characterized by complex, high-volume datasets.

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The author for this study, utilized AI-assisted tools, such as Chat GPT, Copilot, Gemini and Deep Seek, to support the conceptual development, the drafting of the content and specially as a filter towards searching useful information towards the writing of this study. All of the outputs generated by these tools, have been properly verified and properly adapted by the author to ensure accuracy, originality and the overall quality of the content